

Where it is hot and where it is heating are two different lists

Between now and mid-century, which euro-area countries see heat stress grow fastest, and is that ranking the same as the ranking by level of heat stress today?

Why the distinction changes the answer

A book concentrated where the level is already high holds a hazard that should be priced today. One concentrated where the change is largest holds a hazard that arrives inside the maturity of loans written now. If the two orderings agree, today's exposure map still serves. If they do not, it stops serving.

Finding

By level **ESP PRT GRC**

By change **ESP GRC PRT**

The orderings differ: PRT and GRC trade places (page 3).

The gap is small in days; the point is that it is not zero.

At a glance

Indicator	hd35, days $\geq 35^{\circ}\text{C}$
Model	CMIP6 ensemble median
Baseline	1995–2014
Projection	2040–2059
Scenarios	SSP2-4.5, SSP3-7.0
Coverage	11 countries
Source	World Bank CCKP

One indicator, two periods, one subtraction

Source	World Bank Climate Change Knowledge Portal API. Public, no key, no registration, queried live at run time.
Model	CMIP6 ensemble median, 0.25° climatology, aggregated to country.
Indicator	hd35 — days per year with a maximum heat index at or above 35 °C.
Baseline	1995–2014, historical.
Projection	2040–2059, under SSP2-4.5 and SSP3-7.0.
Change	projection minus baseline, per country, in days per year.
Coverage	11 countries: ESP PRT GRC ITA FRA POL AUT DEU BEL NLD IRL.
Requests	one call per collection for the whole country list, not one per country.

What the code is defended against

The API keys each value by a period string, and that key is not the same across collections. Reading by key returned empty baselines, which silently made every change equal to the projection: plausible numbers, wrong ones. `tests/test_hazard.py` now pins that the value is read from the payload rather than from its key, and that change is projection minus baseline and not the reverse.

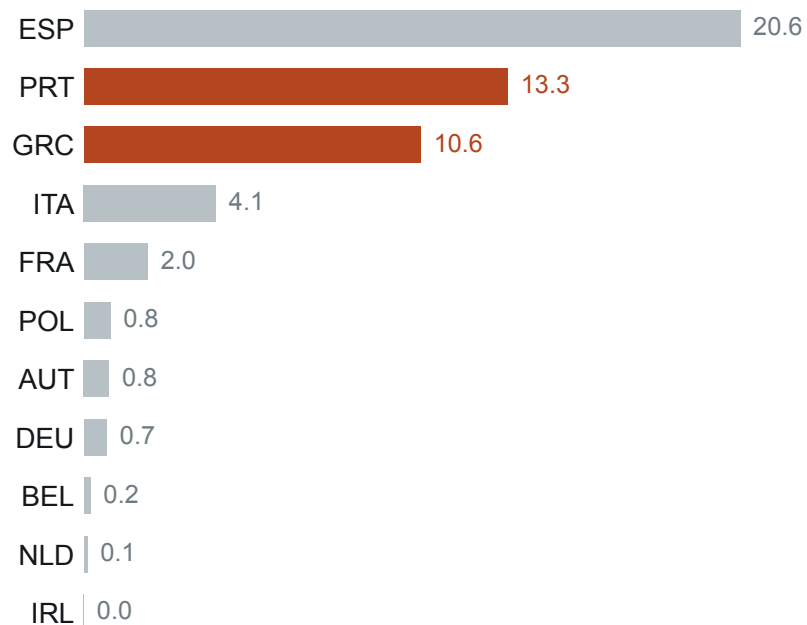
PRT and GRC trade places between level and change

PRT ranks 2nd by level (13.3 days per year) but 3rd by change (+8.2 days).

GRC is the reverse: 3rd by level (10.6 days), 2nd by change (+8.4 days).

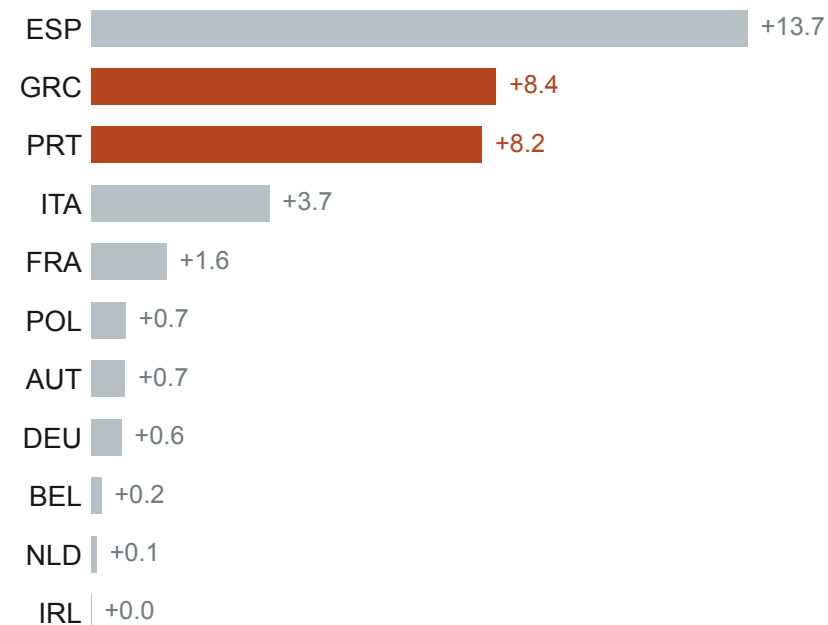
By level, 2040–2059

days per year at or above 35 °C



By change from baseline

additional days per year vs 1995–2014



The left panel is a photograph, the right panel is the direction of travel. A portfolio ranked on either one alone is answering a different question from the one it thinks it is answering.

Source: World Bank Climate Change Knowledge Portal, CMIP6 ensemble median, indicator hd35. Baseline 1995–2014 against 2040–2059, SSP3–7.0.

What this does not tell you

Country means hide the hazard.

Heat stress is urban and local. A national mean over Spain averages Seville with Bilbao. Countries are used because exposure data is reported by country, not because the hazard is national.

Ensemble median only.

No spread, therefore no view on the tail, which is the part a supervisor asks about first. An interquartile range in place of the median is the first thing I would add.

hd35 is one indicator.

It says nothing about drought, flood, wildfire or wind, which reach a balance sheet through different channels and on different timescales.

No exposures.

This is the hazard layer alone: an input to a risk view, not a risk view. Overlaying it is module 02, and doing that honestly needs exposure data at a finer grain than country, which is the real constraint and the reason 01 stops here.

Everything above is a limit on the claim, not a caveat on the code. The numbers are what the model says; what they cannot carry is a view on any single asset, any single city, or the tail.

Source: World Bank Climate Change Knowledge Portal, CMIP6 ensemble median, indicator hd35. Baseline 1995-2014 against 2040-2059, SSP3-7.0.